RISC-V COMPUTER

SystemVerilog to freestanding C on a Basys 3

ARCHITECTURE SHEET / 08 NOV 2026

The custom FPGA CPU executes every demo.

Xilinx Artix-7 XC7A35T | RV32I / ILP32 | No OS, interrupts or hardware multiply/divide

PROCESSOR

Reference: single-cycle at 25 MHz.
Optional: five-stage pipeline at 62.5 MHz.
IF > ID > EX > MEM > WB
Forwarding, load-use stalls and branch flushes.
32 registers; x0 is always zero.
Ordered retirement; ECALL / EBREAK halt.

UNIFIED RAM / 64 KiB

Dual-port block RAM; native CPU interfaces.
0000-CFFF: code, data, BSS, heap (52 KiB)
D000-DFFF: reserved scratch (4 KiB)
E000-FFFF: stack (8 KiB); PC starts at 0.

100 MHz input > MMCM > CPU + 25 MHz pixels

MMIO BUS / 0x8000_0000 — Byte enables + registered device reads

USB UART

115200 baud, 8N1.
16-byte receive FIFO.
Prime output + console.

BOARD + COUNTERS

Buttons, switches, LEDs, hex.
Cycle / instruction counters.
Center button: CPU reset.

OPTIONAL VGA

640 x 480; 25 MHz pixels.
80 x 30 text OR 160 x 120
4-bpp pixels scaled 4 x 4.

TAB5 DEBUG + LAUNCHER / PLANNED

Dedicated packet UART, safe RAM ownership, halt/step, trace and diagnostics remain unimplemented.

DEMONSTRATIONS

Primes: sieve to 10,000; 1,229 primes and measured cycles, instructions and CPI over UART.

Life: two bit-packed 160 x 120 grids; four starting patterns and word-at-a-time rendering.

Snake: 40 x 26 playfield, pixel HUD, direction buttons, pause, score and collision handling.

CLOCKS + EVIDENCE

25 / 62.5 MHz are tested CPU configurations, not a new Fmax sweep. Pixel timing stays fixed.

Reference models, native sanitizers, RV32I images and complete VGA/UART pin checks support local verification.

22 RAMB36 blocks with VGA. Implementation timing and utilization reports accompany the phase records.

ACCEPTANCE STILL PENDING

Physical VGA observation, standalone cold-power demos, photos/video and an independent rebuild are unrun. JTAG configuration is volatile; this sheet does not claim unattended flash boot.

Source of truth: docs/BUILD_PLAN.md | Details: docs/phases/PHASE7_DEMOS.md